SPRING
start.spring.io

Dependencies:
- Spring Web
- Spring Data JPA
- MySQL Driver
- H2 Database

=>Novo Projeto:
restapi\src\main\java\br.com.aula.restapi

1).\RestapiApplication:

```
package br.com.aula.restapi;
import org.springframework.boot.SpringApplication;
import org.springframework.boot.autoconfigure.SpringBootApplication;

@SpringBootApplication
public class RestapiApplication {

  public static void main(String[] args) {
    SpringApplication.run(RestapiApplication.class, args);
  }

}
```

2).\controller
AlunoController:

```
package br.com.aula.restapi.controller;
import br.com.aula.restapi.repository.AlunoRepository;
import br.com.aula.restapi.request.RequestAluno;
import br.com.aula.restapi.service.AlunoService;
import org.springframework.beans.factory.annotation.Autowired;
import org.springframework.http.HttpStatus;
import org.springframework.web.bind.annotation.*;

import java.security.Provider;
import java.util.List;

@RestController
@RequestMapping ("aluno")

public class AlunoController {

   @PostMapping
   @ResponseStatus (HttpStatus.CREATED)
   public void cadastroaluno (@RequestBody RequestAluno aluno){
      System.out.println("Nome: " + aluno.getNome());
```

```
        System.out.println("CPF: " + aluno.getCPF());
        System.out.println("RA: " + aluno.getRA());

        service.save(aluno);

    }
    @Autowired
    private AlunoService service;

    @GetMapping
    public List<RequestAluno> consultaaluno (){

        return service.findall();

    }

}
```

3).\repository
AlunoRepository.java:

```
package br.com.aula.restapi.repository;

import br.com.aula.restapi.request.RequestAluno;
import org.springframework.data.jpa.repository.JpaRepository;
import org.springframework.stereotype.Repository;

@Repository
public interface AlunoRepository extends JpaRepository <RequestAluno, Integer>{
}
```

\request
RequestAluno.java:

```
package br.com.aula.restapi.request;

import javax.persistence.*;

@Entity
@Table (name="alunos")
public class RequestAluno {
    @Id
    @GeneratedValue(strategy = GenerationType.AUTO)
    private Integer id;
    @Column
    private String nome;
    @Column
    private String RA;
    @Column
    private String CPF;

    public String getNome() {
        return nome;
```

```
    }

    public void setNome(String nome) {
        this.nome = nome;
    }

    public String getRA() {
        return RA;
    }

    public void setRA(String RA) {
        this.RA = RA;
    }

    public String getCPF() {
        return CPF;
    }

    public void setCPF(String CPF) {
        this.CPF = CPF;
    }

}
```

4).\service
AlunoService:

```
package br.com.aula.restapi.service;

import br.com.aula.restapi.repository.AlunoRepository;
import br.com.aula.restapi.request.RequestAluno;
import org.springframework.beans.factory.annotation.Autowired;
import org.springframework.stereotype.Repository;
import org.springframework.stereotype.Service;

import javax.annotation.processing.Generated;
import java.util.List;

@Service
public class AlunoService {
    @Autowired
    private AlunoRepository repository;
    public void save(RequestAluno aluno){
        repository.save(aluno);
    }

    public List<RequestAluno> findall (){
        return repository.findAll();

    }
```

}

5) Arquivos de configuração do projeto:

A)restapi\.mvm
maven-wrapper.properties:
distributionUrl=https://repo.maven.apache.org/maven2/org/apache/maven/apache-maven/3.8.4/apache-maven-3.8.4-bin.zip
wrapperUrl=https://repo.maven.apache.org/maven2/org/apache/maven/wrapper/maven-wrapper/3.1.0/maven-wrapper-3.1.0.jar
restapi

B)application.properties:
spring.datasource.url=jdbc:h2:mem:testdb
spring.datasource.driverClassName=org.h2.Driver
spring.datasource.username=sa
spring.datasource.password=password
spring.jpa.database-platform=org.hibernate.dialect.H2Dialect
security.oauth2.client.scope=password

C)pom.xml:
<?xml version="1.0" encoding="UTF-8"?>
<project xmlns="http://maven.apache.org/POM/4.0.0"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://maven.apache.org/POM/4.0.0
https://maven.apache.org/xsd/maven-4.0.0.xsd">
<modelVersion>4.0.0</modelVersion>
<parent>
<groupId>org.springframework.boot</groupId>
<artifactId>spring-boot-starter-parent</artifactId>
<version>2.6.7</version>
<relativePath/> <!-- lookup parent from repository -->
</parent>
<groupId>br.com.aula</groupId>
<artifactId>restapi</artifactId>
<version>0.0.1-SNAPSHOT</version>
<name>restapi</name>
<description>Demo project for Spring Boot</description>
<properties>
<java.version>11</java.version>
</properties>
<dependencies>
<dependency>
<groupId>org.springframework.boot</groupId>
<artifactId>spring-boot-starter-data-jpa</artifactId>
</dependency>
<!-- <dependency>-->
<!-- <groupId>org.springframework.boot</groupId>-->
<!-- <artifactId>spring-boot-starter-security</artifactId>-->

```
<!-- </dependency>-->
<dependency>
<groupId>org.springframework.boot</groupId>
<artifactId>spring-boot-starter-web</artifactId>
</dependency>

<dependency>
<groupId>mysql</groupId>
<artifactId>mysql-connector-java</artifactId>
<scope>runtime</scope>
</dependency>
<dependency>
<groupId>org.springframework.boot</groupId>
<artifactId>spring-boot-starter-test</artifactId>
<scope>test</scope>
</dependency>

<!-- https://mvnrepository.com/artifact/com.h2database/h2 -->
<dependency>
<groupId>com.h2database</groupId>
<artifactId>h2</artifactId>
<version>1.4.199</version>
<scope>runtime</scope>
</dependency>
</dependencies>

<build>
<plugins>
<plugin>
<groupId>org.springframework.boot</groupId>
<artifactId>spring-boot-maven-plugin</artifactId>
</plugin>
</plugins>
</build>

</project>
```

7)POSTMAN: criar collection

A)New collectoin (ex: TesteCadastroAluno)

B)Add Request -> POST=localhost:8080/aluno
Body tab (raw) -> JSON type (rodar depois do item "F"):

```
{
   "nome": "Maria",
   "cpf": "55388811166",
   "ra": "23312"
}
```

C)Add Request -> GET=localhost:8080/aluno

Body tab (none)
Response Body Example (rodar depois do item "F"):


```
[
   {
      "nome": "Pedro",
      "ra": "12312",
      "cpf": "12388811166"
   },
   {
      "nome": "Joao",
      "ra": "33312",
      "cpf": "33388811166"
   },
   {
      "nome": "Maria",
      "ra": "23312",
      "cpf": "55388811166"
   }
]
```


D)Add Request -> POST=localhost:8080/oauth/token
Body tab => Add Keys:
-grant_type:password
-username:user
-password:"log no console do IntelliJ => Using generated security password:512833a6-..."

E)Rodar o item "D", copiar o token ("access_token"), voltar no item "B" na tab Authorization e escolher Type "Bearer Token" e colar o valor do Token no campo "Token".

F)Repetir o item "E" para adicionar o "Token" no item "C".

G)Rodar o item "B" (algumas vezes com valores diferentes no Body da requisição POST).

H)Rodar o item "C" e verificar a lista retornada no Response Body com os dados gravados por persistência em Banco de Dados (h2).